Pandas + Seaborn Advanced Cheat Sheet

PANDAS CORE OPERATIONS

```
import pandas as pd

# Load Data
df = pd.read_csv("file.csv")

# Inspect
df.head()
df.info()
df.describe()
df.shape

# Selection
df["col"]
df[["col1","col2"]]
df.iloc[0:5]
df.loc[0, "col"]

# Filtering
df[df["age"] > 30]
df[(df["age"] > 30) & (df["salary"] > 50000)]

# Missing Values
df.isna().sum()
df.fillna(df.median(numeric_only=True))
df.dropna()

# Grouping
df.groupby("city")["salary"].mean()
df.groupby("dept").agg({"salary":"mean","age":"max"})

# Feature Engineering
df["new"] = df["a"] + df["b"]
df["category"] = df["age"].apply(lambda x: "old" if x>40 else "young")

# Encoding
pd.get_dummies(df)
```

SEABORN ADVANCED VISUALIZATION

```
import seaborn as sns
import matplotlib.pyplot as plt

# Distribution
sns.histplot(df["age"], kde=True)
sns.kdeplot(df["salary"])

# Relationships
sns.scatterplot(x="age", y="salary", data=df)
sns.lineplot(x="year", y="sales", data=df)

# Categorical
sns.barplot(x="dept", y="salary", data=df)
sns.boxplot(x="dept", y="salary", data=df)
sns.violinplot(x="dept", y="salary", data=df)

# Correlation Heatmap
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap="coolwarm")

# Pair Plot
sns.pairplot(df)

# Styling
sns.set(style="whitegrid")
plt.show()
```

PRACTICAL PERFORMANCE TIPS

- Use vectorized operations instead of loops
- Avoid apply() when possible
- Use categorical dtype for memory efficiency

- Drop unused columns early
- Use .copy() to avoid chained assignment issues